

CATCH THE STAR GAME!

A Fun Scratch Jr Coding Adventure

Hey Little Code Explorers!

Have you ever wanted to build your very own video game? Welcome to our ultimate block-based coding guide! In this lesson, we will learn how to command characters, build rules, and fix secret glitches. By using simple code blocks, you will make a game where you jump up and catch bright twinkling stars right on your tablet screen!

Target Grade Level: Kindergarten to Grade 2 (Ages 5-8)

1. THE MAGICAL STARRY NIGHT DREAM

Imagine stepping out onto your open roof during a crisp, beautiful Diwali night! The sky is completely dark, and high above you, millions of brilliant diamonds are twinkling and flashing in the dark.

Have you ever reached your hands up high and wished you could catch a real star? Today, we are going to use our coding superpowers to turn that dream into a real game! Instead of just playing games made by other people, you are going to become a digital architect and design a real game yourself.

2. MEETING YOUR DIGITAL PLAYGROUND

Before we can write instructions for our computer, we need to know where everything lives. The Scratch Jr screen is split into special zones that work exactly like a real puppet theater or school playground!

The 3 Main Zones of Scratch Jr:

1. **The Canvas (The Stage):** This is the big rectangular window where your game comes alive! It is the stage where your characters move, jump, dance, hit boundaries, and show off your code.
2. **The Background Library:** This is the scene room. You click here to pick where your adventure happens! For our game today, we will look for a deep space backdrop or a lovely dark night sky full of clouds.
3. **The Sprite Palette (The Actor House):** This is where all your game characters live. In coding, characters and objects are called *sprites*. This is where we go to cast our actors into the game.

Our Active Game Actor Roster:

Character Role	What Do They Do?	Cool Custom Idea
The Catcher	Slides smoothly back and forth at the bottom of the stage to grab targets.	An astronaut, a kid, or a funny monkey!
The Star Target	Twinkles at the top, vanishes when caught, and teleports to a new spot!	A yellow star or a delicious golden mango!

3. CODING CONCEPT 1: STEP-BY-STEP SEQUENCES

Computers are super fast, but they don't have brains or intuition! They cannot guess what you want them to do. They follow your instructions blindly, one by one, moving from left to right or top to bottom. The exact order of your steps is called a **sequence**.

The Lemonade Rule: Making Nimbu Pani

To understand sequencing, let's think about how we make a glass of delicious, cold Nimbu Pani (Indian lemonade). You have to follow a strict recipe order:

Step ID	Kitchen Action Description	What a Computer Thinks
Step 01	Pour pure water into a tall, clean glass.	Setup: Setting the base state before anything runs.
Step 02	Squeeze a fresh, juicy lemon into the water.	Changing Values: Adding a flavor variable!
Step 03	Mix in sweet sugar and stir it around fast.	Processing: Mixing everything to get the final drink!

The Law of Broken Logic: What happens if you scramble the order? Imagine trying to stir the spoon and squeeze the lemon before pouring any water into the glass! You will end up with dry sugar granules stuck to the bottom and sticky lemon drops spilled all over your counter table. The drink is ruined because you broke the sequence! In coding, if you put blocks in the wrong order, your game breaks exactly the same way.

4. CODING CONCEPT 2: EVENT HANDLERS (TRIGGERS)

When you start a game, why does a character suddenly jump? It's because of an Event! An **Event** is a secret signal or trigger that your code listens for. An **Event Handler** is the special block that wakes up and runs your script when that signal happens.

Real-World Clues for Event Triggers:

- **Example A: Clapping Starts the Song:** Imagine standing in a dark room at a surprise birthday party. Everyone is completely quiet. Suddenly, the leader claps their hands! Instantly, the whole room starts singing "Happy Birthday!" The clap is the *Event Trigger*, and singing is the *Action*.
- **Example B: School Bell Starts Recess:** You are sitting at your school desk focusing hard on your work. The absolute second the high-pitched school bell rings, you drop your pencil and run outside to play on the green grass! The bell ringing is the *Event*, and running outside is the *Action*.

Inside Scratch Jr, the **Green Flag** block is like our birthday clap—it sends a loud signal across the screen saying: "Wake up, start the game!" The **Start on Tap** block is like our school bell—touching the star character triggers the score action!

5. CODING CONCEPT 3: LOOPS (GOING ROUND & ROUND)

Imagine your teacher asked you to write the word "Hello" one hundred times on a paper page. Your hand would get super tired! In computer programming, we use a beautiful shortcut called a **Loop**.

The Loop Rule: Loops repeat actions forever, creating beautiful patterns that never stop!

Two Fun Everyday Loops We Can See:

1. **The Traffic Light Loop:** Watch a busy street crossroads. The electric light alternates in a strict, perfect loop: Red, then Yellow, then Green, and then right back to Red. It loops all day and night to keep cars safe.
2. **The Culinary Jalebi Loop:** Watch a sweet chef cook fresh, hot Jalebis at a local food market stall. The chef doesn't draw broken, straight lines. Instead, they swirl their hand round and round in circles, looping the batter through the hot oil without breaking the stream!

6. SETTING UP YOUR STAR GAME STAGE

Now that we have discovered our three magical coding superpowers (Sequences, Events, and Loops), it is time to build our Catch the Star game project! Grab your tablet and let's follow these instructions together.

STEP 1: PICKING YOUR BACKGROUND SCENE

Click on the landscape picture button at the top of your screen to open the Background Library. Scroll down past the sunny hills and pick the beautiful **Night Sky** backdrop filled with stars, or the deep blue **Outer Space** layout.

STEP 2: CASTING YOUR GAME ACTORS

Go to your left-hand side character panel. Delete the default cat character by pressing and holding it. Now, hit the large plus (+) button to add two brand new sprites:

- **1. Pick Your Catcher Asset:** Look for a character that looks ready to run around at the bottom of your screen. You can choose an adventurous Astronaut, a happy kid explorer, or a cute little Monkey!
- **2. Pick Your Target Asset:** Look for the bright yellow Star sprite. Once it appears on your canvas stage, use your finger to drag it all the way up to the top ceiling edge of the window. This is its starting location.

Creative Personalization Tip: You can theme your game however you want! If you don't want an astronaut catching stars, you can change your catcher to a clever monkey and change your target to a sweet golden **Mango**! Choose assets that make you smile.

7. CODING THE CATCHER: SMOOTH SLIDES!

Click on your Catcher character icon in your palette house. We want to write code that makes him slide smoothly left and right across the bottom of the screen continuously as soon as the game opens up.

The Catcher Block Assembly Recipe:

Go down to your blocks inventory workspace. Look for the colorful block groups and snap these pieces together from left to right like a puzzle train:

Block Shape	Block Category	What It Tells the Catcher to Do
Green Flag	Yellow Triggers	Wakes up the Catcher character the moment the game's main green flag is clicked!
Move Right (5)	Blue Motions	Commands the sprite to march 5 spaces to the right across the stage grid.
Move Left (5)	Blue Motions	Commands the sprite to reverse directions and run 5 steps back to the left side.
Forever Arrow	Red/Orange Control	Wraps tightly around both blue motion arrows, forcing the catcher to slide back and forth nonstop!

8. CODING THE STAR: POP AND TELEPORT!

Now, click on the yellow **Star** character icon. The star needs a very exciting, dynamic code script. It needs to stay still until a finger touches it, make a funny noise, vanish from sight, wait a brief second, and then teleport randomly to a new location!

The Star Target Code Blueprint:

Assemble these puzzle blocks in your workspace grid from left to right:

1. **Start on Tap Block:** This is our touch sensor trigger event handler. It listens for the moment a player clicks or touches the star on the screen.
2. **Play Pop Sound Block:** Snap this right after the touch block to give instant, funny auditory feedback! **Pop!**
3. **Hide Block:** This turns our star completely invisible, making it look like it was successfully caught and tucked away into your score pocket!
4. **Wait Block (2.0s):** Creates a brief 2-second pause so the player can catch their breath and get ready for the next level.
5. **Show at Random Place:** Teleports the star to a brand new, sneaky, unpredictable spot on the canvas screen and turns it bright and visible again!

9. LEVEL UP: THREE MINI-CHALLENGES!

Once your basic star game is running smoothly, it's time to upgrade your code script! Try out these three advanced micro-challenges to explore new hidden block sets:

- **CHALLENGE 1: THE KINETIC ROTATION SPIN:** Can you make the star spin around like a dizzy top the exact moment the catcher grabs it? Go to your blue block group, look for the circular **Rotate Block**, and insert it right after the touch trigger block!
- **CHALLENGE 2: DYNAMIC SCALE GROWTH:** Let's make the game harder! Can you program the star to grow larger and larger every time it respawns? Find the purple **Grow Block** inside your visibility look menu and snap it into your code script.
- **CHALLENGE 3: THEME SWAP EXERCISE:** Completely transform the style of your game! Head back into the asset library and swap out your astronaut for an animal, or trade the star for a piece of fruit, proving you know how to navigate the library maps.

10. BECOMING A SOFTWARE GLITCH DETECTIVE!

In the world of computer engineering, a mistake or error in code is called a **Bug**. Tracking down those mistakes and fixing them is called **Debugging**! Making mistakes is a normal part of science, and debugging is a fantastic way to build a strong, patient brain.

Two Common Game Bugs & How to Squash Them:

Bug 1: The Frozen Sprite Glitch

The Problem: You click the big green flag, but your Catcher sprite stands completely frozen like an ice statue! He won't move at all.

The Fix: Look at your script line area. Make sure your yellow **Green Flag Trigger** block is physically snapped and locked onto the front of your blue motion blocks. If it is floating nearby without touching, the train won't start running!

Bug 2: The Vanishing Star Mystery

The Problem: You touch the star once, it disappears into thin air, and it never, ever comes back to the screen! It is gone forever.

The Fix: Look at the tail end of your star's code train. Many students attach a purple **Hide Block**, but they forget to attach the final purple **Show Block** at the very end! The code is working, but your poor star is stuck in an invisible ghost state!

11. YOUR ULTIMATE CODE RECALL REVIEW

Outstanding job, Software Architects! You have built a complete, interactive video game from scratch. Let's review our three golden coding rules before we head home:

1. **Sequence:** Step-by-step instructions that must follow a perfect, strict order—just like mixing our sweet Nimbu Pani. If you scramble the recipe order, the computer code collapses!
2. **Events:** A trigger signal (like a screen touch or finger click) that tells a script block to wake up and run—just like a loud school bell starting recess!
3. **Loops:** Magical control blocks that repeat an action continuously without ever stopping—just like a traffic light switching colors or sweet jalebi batter swirling in a frying pot!

Your Home Sandbox Mission:

Open Scratch Jr on your tablet at home with your parents. Design a brand new standalone game using two characters you haven't tried yet! Your home project must include at least one continuous moving loop and use a finger-touch event trigger block. Invite your parents to test your game and see if they can catch your targets! Keep exploring, code stars! 🏆